

WASP-52b

R-1125

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_211116 · created 2026-07-30T22:23:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459534.72 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.044
Transit depth (Rp/Rs)^2	2.73 +/- 1.46 [%]
Orbital Inclination (inc)	84.15 +/- 2.06
Airmass coefficient 1 (a1)	0.873 +/- 0.021
Airmass coefficient 2 (a2)	0.089 +/- 0.014
Scatter in the residuals of the lightcurve fit is	2.42 %
Best Comparison Star	#1 - [521, 286]
Optimal Method	PSF photometry
Transit Duration (day)	0.061 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.165 ± 0.044 vs 0.1646 (deviation 0.01σ)
Duration fit vs published	0.061 d vs 0.0758 d (deviation 0.4σ)
Mid-time O–C	9.1 min
Depth SNR	2.3
Residual scatter	2.42 %

Light curve

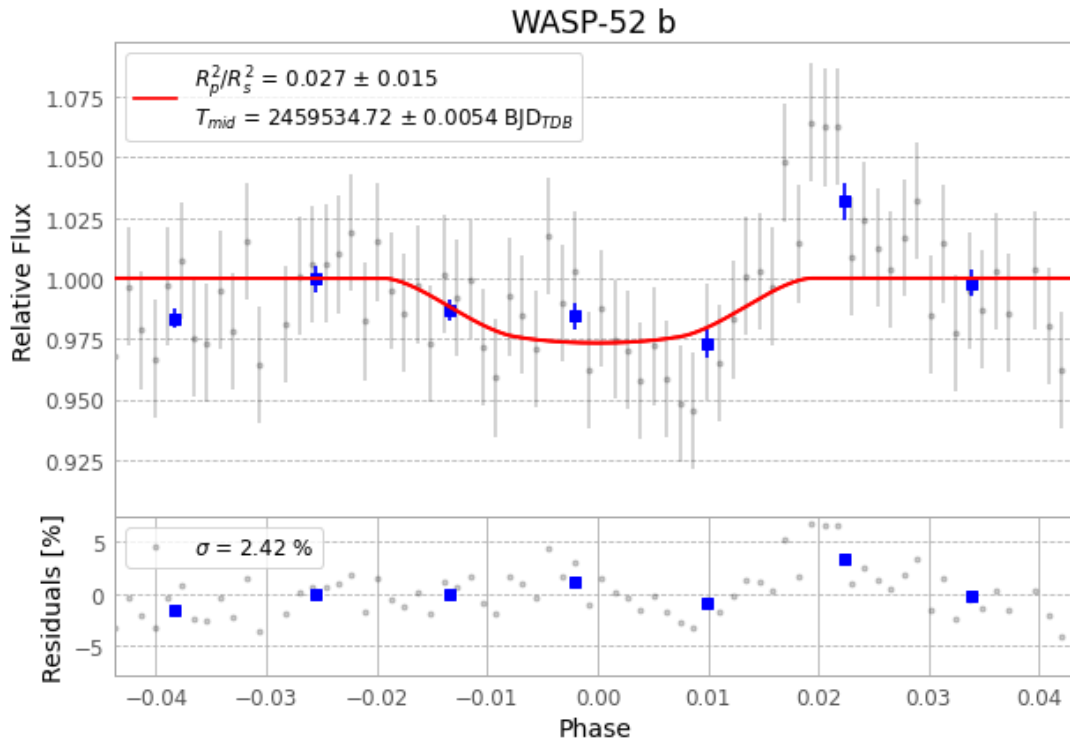


Figure 1 — FinalLightCurve_WASP-52 b_2021-11-15.png (SHA-256 e955fe66e3d9f108...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [403, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fd932e3ac45ed506...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-52211116032112.FITS → WASP-52211116070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-211116.log (ae32ba545afa...), FinalLightCurve_WASP-52 b_2021-11-15.png (e955fe66e3d9...), FinalLightCurve_WASP-52 b_2021-11-15.csv (627aa6387ad6...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 22:23Z.